

Data Interpretation Exercise

Data Interpretation Assessment (R Programming)

Question 1: Mapping Data onto Aesthetics

Dataset

Product	Sales (Lakhs ₹)	Profit (Lakhs ₹)	Category
P1	120	18	Electronics
P2	90	12	Furniture
P3	150	25	Electronics
P4	80	8	Furniture
P5	135	22	Electronics
P6	100	16	Furniture

1. R Program

```
# Load library
```

```
library(ggplot2)
```

```
# Create dataset
```

```
data <- data.frame(
```

```
  Product = c("P1","P2","P3","P4","P5","P6"),
```

```
  Sales = c(120,90,150,80,135,100),
```

```
  Profit = c(18,12,25,8,22,16),
```

```
  Category = c("Electronics","Furniture",
```

```
               "Electronics","Furniture",
```

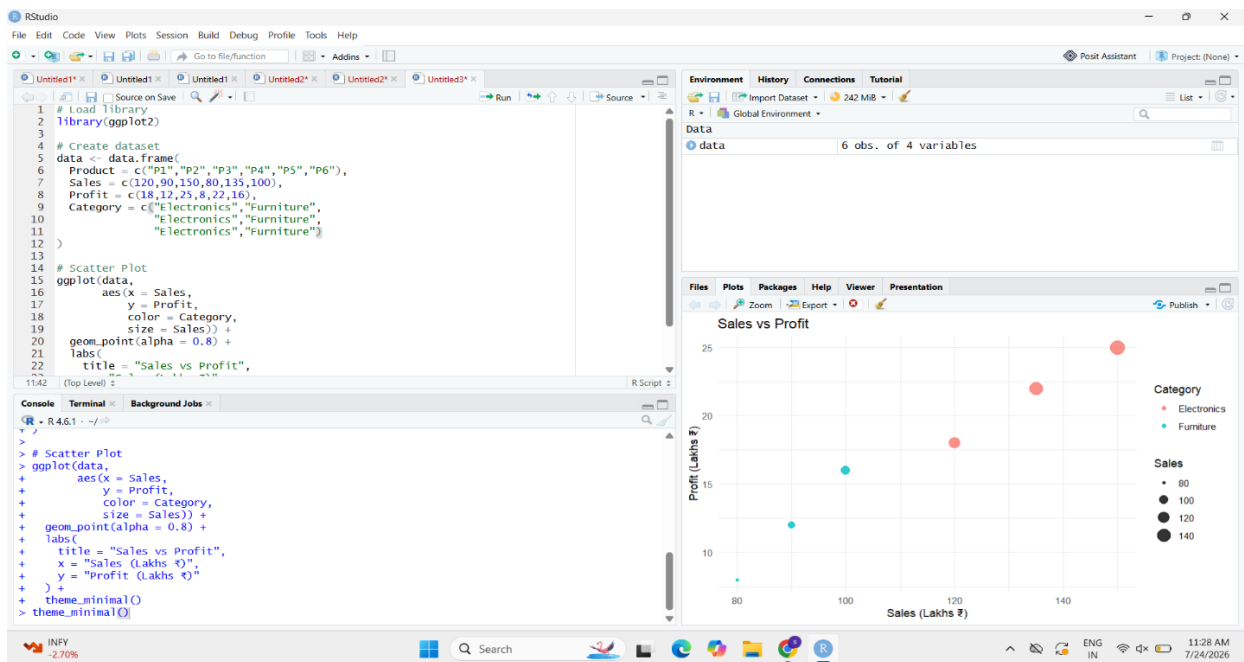
```
               "Electronics","Furniture")
```

)

Scatter Plot

```
ggplot(data,  
  aes(x = Sales,  
      y = Profit,  
      color = Category,  
      size = Sales)) +  
geom_point(alpha = 0.8) +  
labs(  
  title = "Sales vs Profit",  
  x = "Sales (Lakhs ₹)",  
  y = "Profit (Lakhs ₹)"  
) +  
theme_minimal()
```

Graph



2. Map Category using an appropriate aesthetic and Sales using another suitable aesthetic

In the scatter plot:

- **Category** is mapped to the **colour aesthetic**. Different colours represent different product categories (Electronics and Furniture), making it easy to distinguish between them.
- **Sales** is mapped to the **size aesthetic**. Products with higher sales appear as larger points, while products with lower sales appear as smaller points.

Using multiple aesthetics allows the graph to display several variables simultaneously without making it difficult to understand.

3. Interpret the relationship between Sales and Profit

The scatter plot indicates a **positive relationship** between Sales and Profit.

- As **Sales increase**, **Profit also tends to increase**.
- Product **P3** has the highest sales (₹150 lakhs) and also the highest profit (₹25 lakhs).
- Product **P4** has the lowest sales (₹80 lakhs) and the lowest profit (₹8 lakhs).
- The data points follow an upward trend, indicating that higher sales generally lead to higher profits.

Therefore, the visualization suggests a **strong positive correlation** between Sales and Profit.

4. Identify the type of scale used for each variable

Variable	Scale Type	Reason
Sales	Continuous Scale	Numerical values measured in lakhs.
Profit	Continuous Scale	Numerical values that can take different values.
Category	Categorical (Discrete) Scale	Represents different product groups.
Product	Categorical Scale	Product names are labels, not numerical measurements.

5. Suggest one alternative visualization that better represents the same data

A **bubble chart** is an excellent alternative visualization.

Reason:

- It shows the relationship between Sales and Profit.
- Bubble size can represent another variable, such as Profit or Sales.
- Bubble colour can represent Category.
- This enables viewers to compare multiple variables in a single visualization, making it more informative than a basic scatter plot.

Question 2: Coordinate Systems (Cartesian and Polar)

Dataset

Direction	Frequency
N	18
NE	12
E	20
SE	15
S	22
SW	11
W	16
NW	14

1. Write an R program to visualize the dataset using a Cartesian coordinate system.

```
# Load library
```

```
library(ggplot2)
```

```
# Create dataset
```

```
direction <- data.frame(
```

```
  Direction = c("N","NE","E","SE","S","SW","W","NW"),
```

```
  Frequency = c(18,12,20,15,22,11,16,14)
```

```
)
```

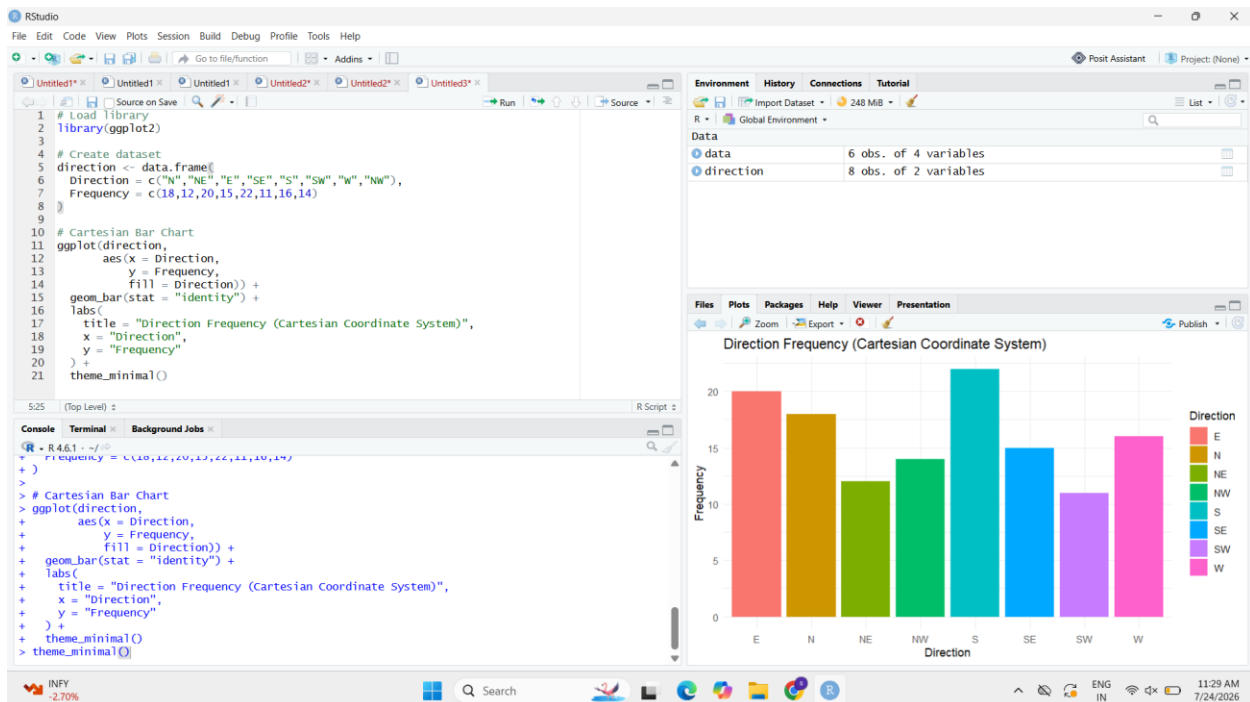
```
# Cartesian Bar Chart
```

```

ggplot(direction,
  aes(x = Direction,
      y = Frequency,
      fill = Direction)) +
geom_bar(stat = "identity") +
labs(
  title = "Direction Frequency (Cartesian Coordinate System)",
  x = "Direction",
  y = "Frequency"
) +
theme_minimal()

```

Graph



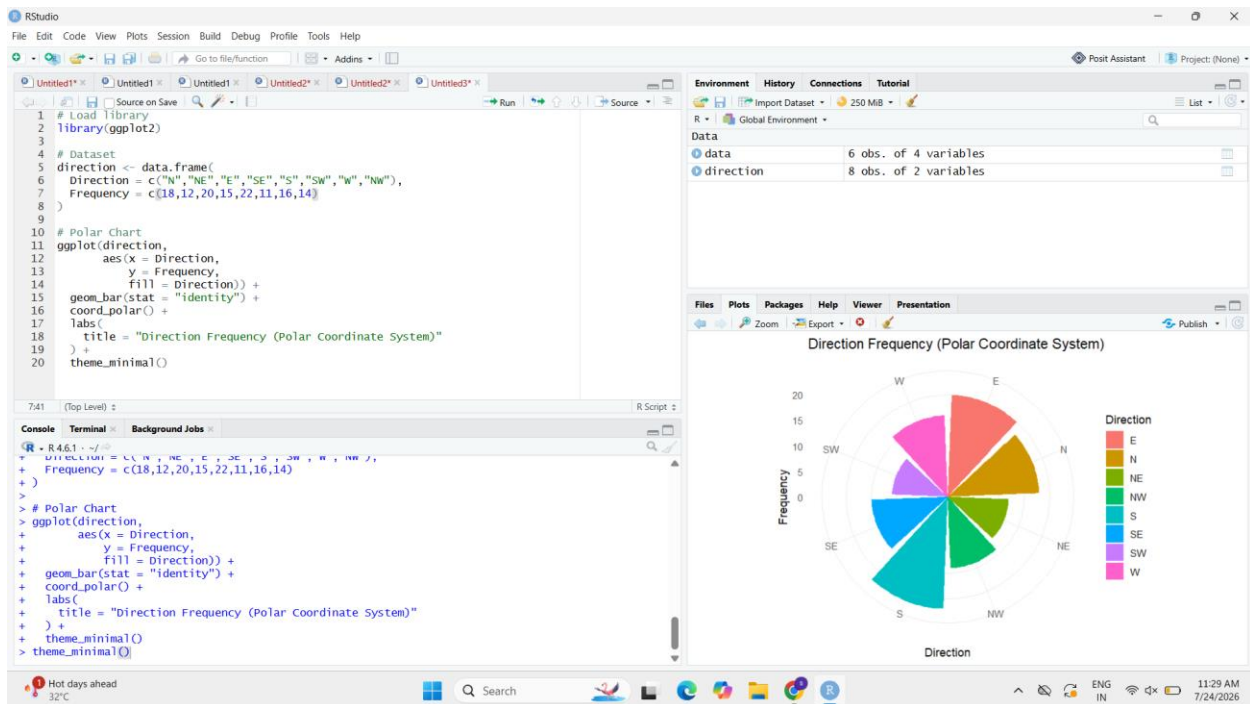
2. Write another R program to visualize the same data using a Polar coordinate system.

```
# Load library
library(ggplot2)

# Dataset
direction <- data.frame(
  Direction = c("N","NE","E","SE","S","SW","W","NW"),
  Frequency = c(18,12,20,15,22,11,16,14)
)

# Polar Chart
ggplot(direction,
  aes(x = Direction,
      y = Frequency,
      fill = Direction)) +
  geom_bar(stat = "identity") +
  coord_polar() +
  labs(
    title = "Direction Frequency (Polar Coordinate System)"
  ) +
  theme_minimal()
```

Graph



3. Compare the effectiveness of both visualizations.

Cartesian Coordinate System

- Uses horizontal and vertical axes.
- Makes it easy to compare frequencies accurately.
- Differences between values can be identified quickly.
- Suitable when exact numerical comparison is important.

Polar Coordinate System

- Displays data in a circular format.
- Represents directional information naturally.
- Makes the relationship between compass directions easier to understand.
- Better for identifying patterns around a circle than exact numerical differences.

Comparison:

The Cartesian chart is more suitable for comparing values precisely, whereas the Polar chart provides a more intuitive representation of directional data.

4. Explain which coordinate system is more appropriate for directional data.

The **Polar Coordinate System** is more appropriate for directional data because directions such as North, South, East, and West naturally represent angles around a circle. The circular arrangement matches the way directional information is interpreted in real-world applications like navigation, wind direction analysis, weather forecasting, and compass readings.

By displaying the data in a circular layout, the polar chart makes directional relationships easier to understand than a traditional Cartesian chart.

5. Justify your answer based on data interpretation.

The Polar coordinate system is preferred because:

- It represents compass directions in their natural circular order.
- It helps viewers recognize directional patterns more easily.
- Adjacent directions (such as N, NE, and E) appear next to each other, improving interpretation.
- It provides a more realistic representation of directional measurements.

However, if the goal is to compare exact frequencies, the Cartesian coordinate system is more effective because the heights of the bars are easier to compare accurately.

Question 3: Color Scales for Data Representation

Dataset

Student	Marks
A	45
B	58
C	72
D	81
E	67
F	90

1. Write an R program to create a bar chart using a sequential color scale based on Marks.

```
# Load library
```

```
library(ggplot2)
```

```
# Create dataset
```

```
student <- data.frame(  
  Student = c("A","B","C","D","E","F"),  
  Marks = c(45,58,72,81,67,90)  
)
```

```
# Bar chart with sequential colour scale
```

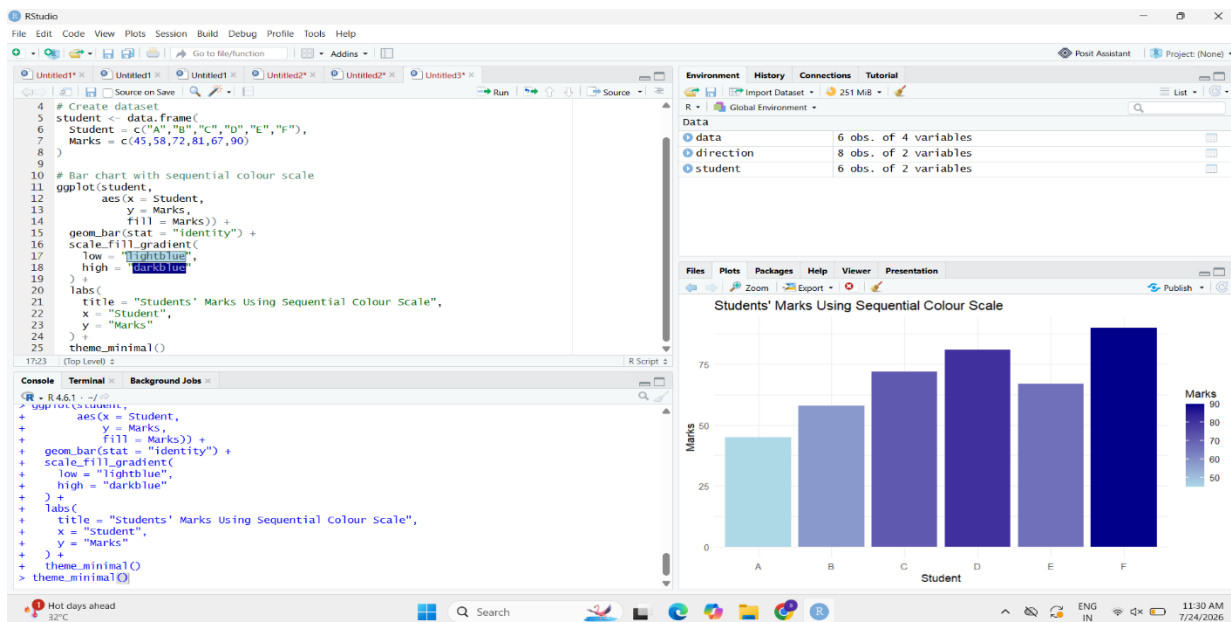
```
ggplot(student,  
  aes(x = Student,
```

```

    y = Marks,
    fill = Marks)) +
geom_bar(stat = "identity") +
scale_fill_gradient(
  low = "lightblue",
  high = "darkblue"
) +
labs(
  title = "Students' Marks Using Sequential Colour Scale",
  x = "Student",
  y = "Marks"
) +
theme_minimal()

```

Graph



2. Identify the students with the highest and lowest marks from the visualization.

From the bar chart:

- **Student F** scored the **highest marks**, which is **90**.
- **Student A** scored the **lowest marks**, which is **45**.

The sequential colour scale also helps identify these students easily. Student F appears with the darkest shade, while Student A appears with the lightest shade.

3. Explain why a sequential colour scale is appropriate.

A sequential colour scale is suitable because **Marks** is a **continuous numerical variable**. Sequential colour scales use a gradual change in colour intensity to represent increasing or decreasing values.

In this visualization:

- Lower marks are displayed using **lighter shades**.
- Higher marks are displayed using **darker shades**.
- The gradual colour transition makes it easy to compare students' performance.
- It improves readability by showing the magnitude of values without requiring users to read every number.

Therefore, a sequential colour scale effectively communicates variations in marks.

4. Suggest a colour palette suitable for colour-blind users.

A colour-blind-friendly palette such as **Viridis** is highly recommended because:

- It is easily distinguishable by most people with colour vision deficiencies.
- It provides a smooth transition from low to high values.
- It maintains good contrast even when printed in grayscale.
- It is widely used in scientific and statistical visualizations.

In **ggplot2**, it can be used as:

`scale_fill_viridis_c()`

5. Explain the difference between sequential and categorical colour scales.

Sequential Colour Scale	Categorical Colour Scale
Used for continuous numerical data.	Used for categorical or discrete data.
Colours change gradually from light to dark.	Each category receives a distinct colour.
Shows magnitude or intensity of values.	Distinguishes different groups without implying order.
Example: Marks, Sales, Temperature.	Example: Gender, Department, Product Category.

Example:

- If the data represents **marks**, a sequential colour scale is appropriate because marks have an ordered numerical range.
- If the data represents **departments** such as CSE, AI&DS, ECE, and IT, a categorical colour scale should be used because these categories have no numerical order.

Question 4: Highlighting with Color

Dataset

Region	Revenue (₹ Crores)
North	820
South	760
East	790
West	1010
Central	770

1. Write an R program to create a bar chart highlighting the region with the highest revenue.

```
# Load library
```

```
library(ggplot2)
```

```
# Create dataset
```

```
region <- data.frame(
```

```
  Region = c("North","South","East","West","Central"),
```

```
  Revenue = c(820,760,790,1010,770)
```

```
)
```

```
# Highlight the highest revenue region
```

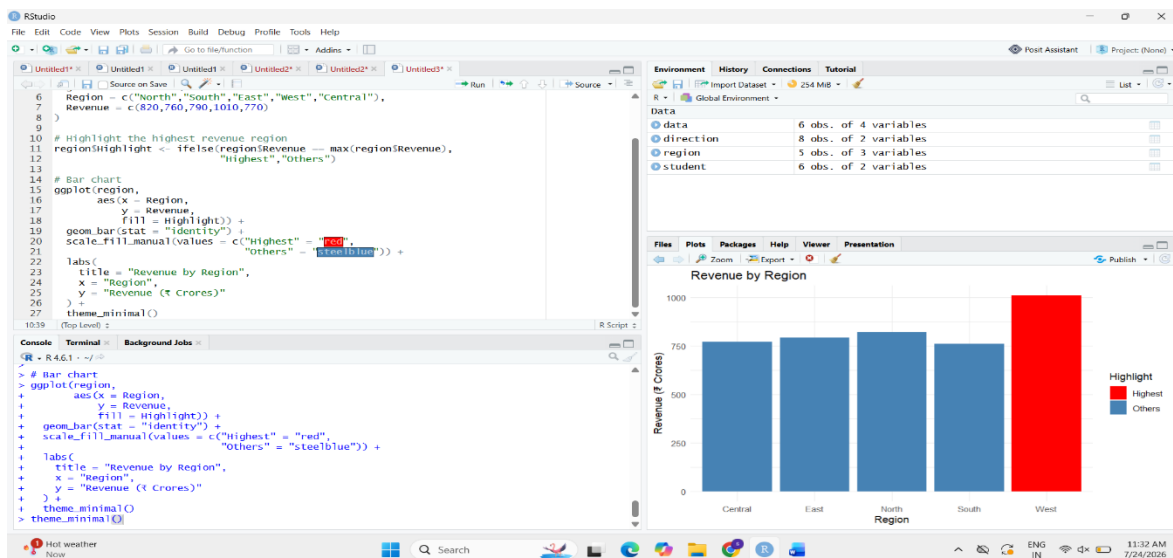
```
region$Highlight <- ifelse(region$Revenue == max(region$Revenue),  
                           "Highest","Others")
```

```

# Bar chart
ggplot(region,
  aes(x = Region,
      y = Revenue,
      fill = Highlight)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c("Highest" = "red",
                              "Others" = "steelblue")) +
  labs(
    title = "Revenue by Region",
    x = "Region",
    y = "Revenue (₹ Crores)"
  ) +
  theme_minimal()

```

Graph



2. Explain how highlighting improves data interpretation.

Highlighting is a visualization technique used to draw the viewer's attention to the most important information in a graph. In this chart, the **West** region is displayed in a different colour (red), while the other regions are shown in a common colour (blue).

The benefits of highlighting include:

- It immediately attracts the viewer's attention to the highest-performing region.
- It makes comparisons easier without reading every value.
- It emphasizes key findings and supports faster decision-making.
- It improves the readability of the graph by separating important information from less important data.

Thus, highlighting enhances the effectiveness of data visualization and helps users identify significant patterns quickly.

3. Identify the highest-performing region.

From the given dataset:

Region Revenue (₹ Crores)

North 820

South 760

East 790

West 1010

Central 770

The **West** region is the **highest-performing region**, with a revenue of **₹1010 crores**.

4. Discuss how excessive use of colours can affect interpretation.

Although colours improve visualization, using too many colours can reduce clarity.

Some disadvantages of excessive colour usage are:

- It makes the graph look cluttered and confusing.
- Important information may become difficult to identify.
- Similar colours may be mistaken for one another.
- It distracts the viewer from the main message.
- It creates problems for people with colour vision deficiencies.

Therefore, colours should be used carefully and only to highlight meaningful information.

5. Suggest one alternative highlighting technique other than colour.

An effective alternative highlighting technique is **data labels**.

Instead of changing the colour, the highest bar can be labeled with its exact value (e.g., **1010**) or a label such as "**Highest Revenue**".

Other alternatives include:

- Using a different bar pattern (striped or dotted).
- Increasing the width of the important bar.
- Adding an annotation or arrow pointing to the highest value.
- Making the important bar bold with a thicker border.

These methods effectively highlight important information even without relying on colour.

Question 5: Integrated Visualization (Bubble Chart)

Dataset

City	Population (Million)	AQI	Rainfall (mm)	Region
A	8.2	110	850	North
B	5.6	95	920	South
C	9.4	140	780	East
D	6.8	105	890	West
E	4.9	88	950	Central

1. Write an R program to create a bubble chart using appropriate aesthetic mappings.

```
# Load library
```

```
library(ggplot2)
```

```
# Create dataset
```

```
city <- data.frame(
```

```
  City = c("A","B","C","D","E"),
```

```
  Population = c(8.2,5.6,9.4,6.8,4.9),
```

```
  AQI = c(110,95,140,105,88),
```

```
  Rainfall = c(850,920,780,890,950),
```

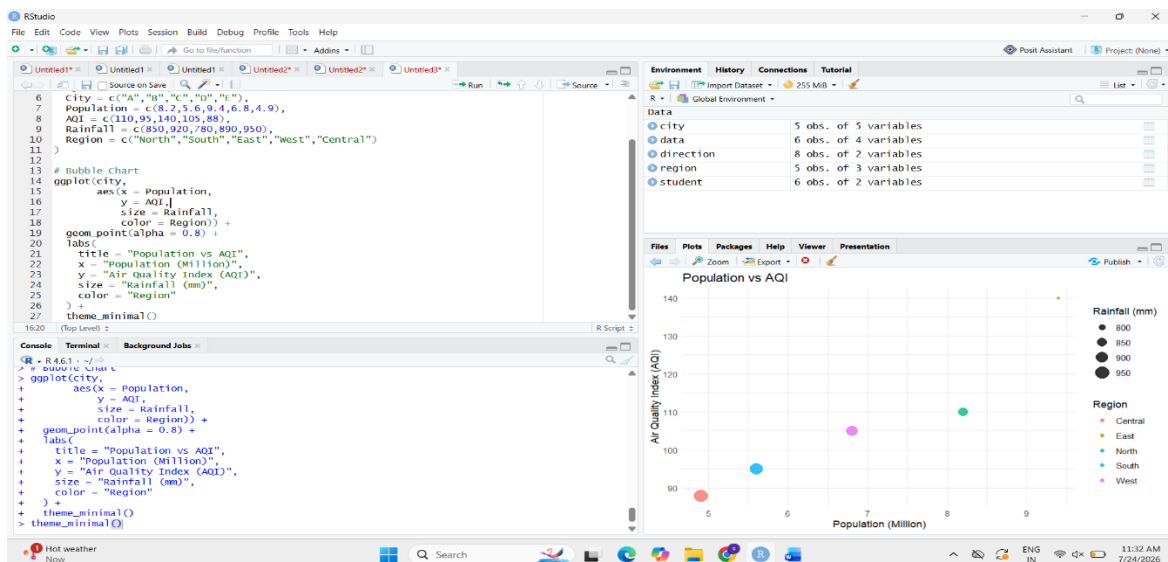
```
  Region = c("North","South","East","West","Central")
```

```
)
```

```
# Bubble Chart
```

```
ggplot(city,
  aes(x = Population,
    y = AQI,
    size = Rainfall,
    color = Region)) +
geom_point(alpha = 0.8) +
labs(
  title = "Population vs AQI",
  x = "Population (Million)",
  y = "Air Quality Index (AQI)",
  size = "Rainfall (mm)",
  color = "Region"
) +
theme_minimal()
```

Chart



2. Select suitable aesthetics for Population, AQI, Rainfall, and Region.

The bubble chart uses different aesthetics to represent multiple variables simultaneously.

Variable	Aesthetic Used	Reason
Population	X-axis	Represents the independent numerical variable.
AQI	Y-axis	Represents the air quality level.
Rainfall	Bubble Size	Larger bubbles indicate higher rainfall.
Region	Colour	Different colours distinguish different regions.

Using these aesthetics enables the visualization of four variables in a single chart, making it informative and easy to interpret.

3. Interpret the relationship between Population and AQI.

The bubble chart shows a **positive relationship** between **Population** and **AQI**.

- Cities with larger populations generally have higher AQI values.
- **City C** has the highest population (9.4 million) and also the highest AQI (140), indicating poorer air quality.
- **City E** has the lowest population (4.9 million) and the lowest AQI (88), indicating comparatively better air quality.
- This suggests that increased population may contribute to higher pollution levels due to factors such as traffic, industries, and urbanization.

Although the dataset is small, the visualization indicates a positive trend between population growth and air pollution.

4. Identify the continuous and categorical variables.

Variable	Type
Population	Continuous
AQI	Continuous
Rainfall	Continuous
Region	Categorical
City	Categorical

Explanation

- **Continuous variables** are numerical values that can be measured on a scale, such as Population, AQI, and Rainfall.
- **Categorical variables** represent labels or groups, such as City names and Regions.

5. Recommend one improvement to enhance the visualization's readability.

One effective improvement is to **add labels for each bubble using the city names**.

For example:

```
geom_text(aes(label = City),  
          vjust = -1,  
          size = 4)
```

This allows viewers to identify each city directly without referring to a legend.

Other improvements include:

- Using a colour-blind-friendly palette (e.g., **Viridis**).

- Adding grid lines for easier comparison.
- Increasing font size for titles and axis labels.
- Adjusting bubble transparency to reduce overlap.
- Sorting data to make comparisons clearer.

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